



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Applied DevOps

Experiment 5

MTECH - School of Computer Science
CYBER SECURITY AND FORENSICS

Submitted To: Prof. KrishnaRaj

Name: Jigesh Sheoran

SAP ID: 590025428

Objective — To explore Git as a DVCS and perform operations like branching, merging, collaboration, and version control management.

Problem Statement:

To perform a series of tasks to explore **Git as a Distributed Version Control System (DVCS)** and demonstrate its advantages over centralized version control systems. Execute all steps carefully and record your observations wherever required.

Tools and Environment :

Development Environment

- Operating System: macOS (MacBook Air)
IDE: Visual Studio Code
- Terminal: zsh shell

Version Control

- Git: Used for tracking source code changes and managing versions
- GitHub: Cloud-based repository hosting platform
Authentication Method: SSH key-based authentication.

Task to be Performed

Task 1: Repository Initialization

1. Create a new directory named dvcs-lab.
2. Initialize it as a Git repository.
3. Create a file file.txt and add initial content.
4. Stage and commit the file with an appropriate message.

Deliverable:

```
dvcs-lab -- -zsh -- 129x38

/Users/sheoraninfosec
sheoraninfosec@Jigeshs-MacBook-Air ~ % ls
Applications
Creative Cloud Files 500087265@stu.upes.ac.in 0efd2be8340bd3b1f8aec782ef0cc2099fd5df328bf375f115a125577e95d625
Creative Cloud Files Personal Account jigeshsheoran2002@gmail.com 4EB975E15F4FED470A495EAE@AdobeID
Desktop
devops-lab-demo
Documents
Downloads
dvcs-lab
iCloud Drive (Archive)
lambda-exp
Library
misp-docker
Movies
Music
ns2-docker
Pictures
Public
Virtual Machines.localized
sheoraninfosec@Jigeshs-MacBook-Air ~ % cd dvcs-lab
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/sheoraninfosec/dvcs-lab/.git/
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % clear
```

```
dvcs-lab -- -zsh -- 129x38

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % echo "Devops Lab Experiment 5" > file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git add file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git commit -m "Initial commit"
[master (root-commit) f4048e7] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %
```

Task 2: Demonstrating Distributed Nature

1. Clone the repository into a new directory named dvcs-lab-clone.
2. Verify that the cloned repository contains the full commit history.
3. Make changes in the cloned repository and commit them **without connecting to any remote server**.
4. Explanation of how this task demonstrates DVCS

Explanation:

- Clone contains entire commit history, not just latest files
- You can commit without internet or server
- Unlike centralized systems, no dependency on a central server

```
dvcs-lab-clone — zsh — 129x38

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % cd ..
[sheoraninfosec@Jigeshs-MacBook-Air ~ % git clone dvcs-lab dvcs-lab-clone
Cloning into 'dvcs-lab-clone'...
done.
[sheoraninfosec@Jigeshs-MacBook-Air ~ % cd dvcs-lab-clone
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git log
commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78 (HEAD -> master, origin/master, origin/HEAD)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:40:37 2026 +0530

Initial commit
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % echo "Change from clone" >> file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git add file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git commit -m "Update from clone"
[master 8d523ac] Update from clone
1 file changed, 1 insertion(+)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git log
commit 8d523ac1b911d0fcae0748e9ff47769e255e176 (HEAD -> master)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:49:53 2026 +0530

Update from clone

commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78 (origin/master, origin/HEAD)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:40:37 2026 +0530

Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone %
```

Task 3: Branching and Merging

1. Create a new branch named feature-branch.
2. Switch to the branch and modify file.txt.
3. Commit the changes.
1. Switch back to the main branch and merge the feature branch.

Branching is efficient in git because:

- Branching is lightweight (pointer-based) → instant creation
- Multiple features can be developed in parallel
- Efficient because Git does not duplicate files

```
dvcs-lab — -zsh — 129x38

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % cd ../dvcs-lab
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git branch feature-branch
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git checkout feature-branch
Switched to branch 'feature-branch'
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % echo "Feature work" >> file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git add file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git commit -m "Added feature"
[feature-branch d95bd79] Added feature
 1 file changed, 1 insertion(+)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git checkout main
error: pathspec 'main' did not match any file(s) known to git
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git checkout master
Switched to branch 'master'
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git merge feature-branch
Updating f4048e7..d95bd79
Fast-forward
 file.txt | 1 +
 1 file changed, 1 insertion(+)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log
commit d95bd79a1cbb9225d1946ef980e5cbd70da66b7 (HEAD -> master, feature-branch)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date:   Wed Mar 25 09:55:27 2026 +0530

    Added feature

commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date:   Wed Mar 25 09:40:37 2026 +0530

    Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %
```

Task 4: Simulating Collaboration

1. Add a remote repository (e.g., GitHub).
2. Push your local repository to the remote.
3. Pull the changes into the cloned repository.

Setting up SSH key:

```
dvcs-lab — zsh — 129x38

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git remote add origin https://github.com/sheoraninfosec/dvcs-lab.git
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/sheoraninfosec/dvcs-lab.git'
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git push -u origin master
Username for 'https://github.com': sheoraninfosec
Password for 'https://sheoraninfosec@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/sheoraninfosec/dvcs-lab.git/'
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % ls -al ~/.ssh
total 48
drwx----- 8 sheoraninfosec staff 256 22 Jan 12:16 .
drwxr-x---- 44 sheoraninfosec staff 1408 25 Mar 09:56 ..
-rw----- 1 sheoraninfosec staff 419 22 Jan 12:16 id_ed25519
-rw-r--r-- 1 sheoraninfosec staff 104 22 Jan 12:16 id_ed25519.pub
-rw----- 1 sheoraninfosec staff 2675 1 Nov 2023 id_rsa
-rw-r--r-- 1 sheoraninfosec staff 594 1 Nov 2023 id_rsa.pub
-rw----- 1 sheoraninfosec staff 828 1 Nov 2023 known_hosts
-rw-r--r-- 1 sheoraninfosec staff 92 1 Nov 2023 known_hosts.old
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % eval "$(ssh-agent -s)"
Agent pid 89955
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % ssh-add ~/.ssh/id_ed25519
Identity added: /Users/sheoraninfosec/.ssh/id_ed25519 (jsheoran9755@gmail.com)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % pbcopy < ~/.ssh/id_ed25519.pub
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % ssh -T git@github.com
Hi sheoraninfosec! You've successfully authenticated, but GitHub does not provide shell access.
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %
```

```
dvcs-lab-clone — zsh — 129x50

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git remote -v
origin https://github.com/sheoraninfosec/dvcs-lab.git (fetch)
origin https://github.com/sheoraninfosec/dvcs-lab.git (push)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git remote set-url origin git@github.com:sheoraninfosec/dvcs-lab.git
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git remote -v
origin git@github.com:sheoraninfosec/dvcs-lab.git (fetch)
origin git@github.com:sheoraninfosec/dvcs-lab.git (push)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git push -u origin master
ERROR: Repository not found.
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 488 bytes | 488.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/dvcs-lab.git
 * [new branch] master -> master
branch 'master' set up to track 'origin/master'.
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % cd ../dvcs-lab-clone
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git pull origin main
fatal: couldn't find remote ref main
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 262 bytes | 131.00 KiB/s, done.
From /Users/sheoraninfosec/dvcs-lab
 * branch master -> FETCH_HEAD
 f4048e7..d95bd79 master -> origin/master
hint: You have divergent branches and need to specify how to reconcile them.
hint: You can do so by running one of the following commands sometime before
hint: your next pull:
hint:
hint: git config pull.rebase false # merge
hint: git config pull.rebase true # rebase
hint: git config pull.ff only # fast-forward only
hint:
hint: You can replace "git config" with "git config --global" to set a default
hint: preference for all repositories. You can also pass --rebase, --no-rebase,
hint: or --ff-only on the command line to override the configured default per
hint: invocation.
fatal: Need to specify how to reconcile divergent branches.
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone %
```

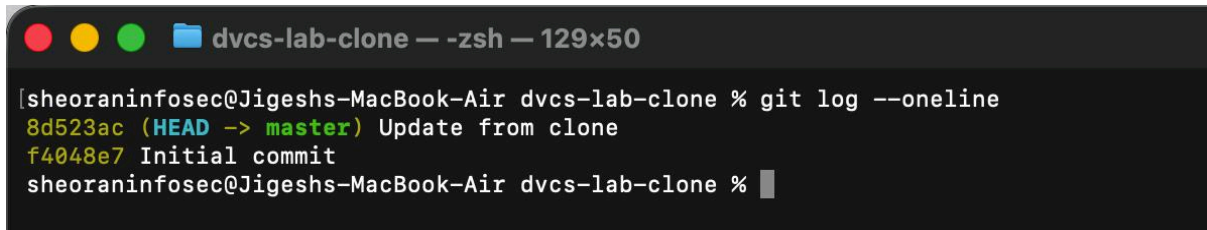
Explanation

- **git push** → uploads local commits to remote repository
- **git pull** → fetch + merge updates from remote
- This simulates real-world team collaboration

Task 5: Exploring Commit History

1. Display the commit history in a compact format.
2. Analyze the sequence of commits.

Deliverable:

A terminal window titled 'dvcs-lab-clone — zsh — 129x50' showing the output of the command 'git log --oneline'. The output lists two commits: '8d523ac (HEAD -> master) Update from clone' and 'f4048e7 Initial commit'. The prompt is 'sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone %'.

```
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone % git log --oneline
8d523ac (HEAD -> master) Update from clone
f4048e7 Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab-clone %
```

Explanation

- Shows commit history in compact form
- Helps track:
 - Changes
 - Authors
 - Timeline

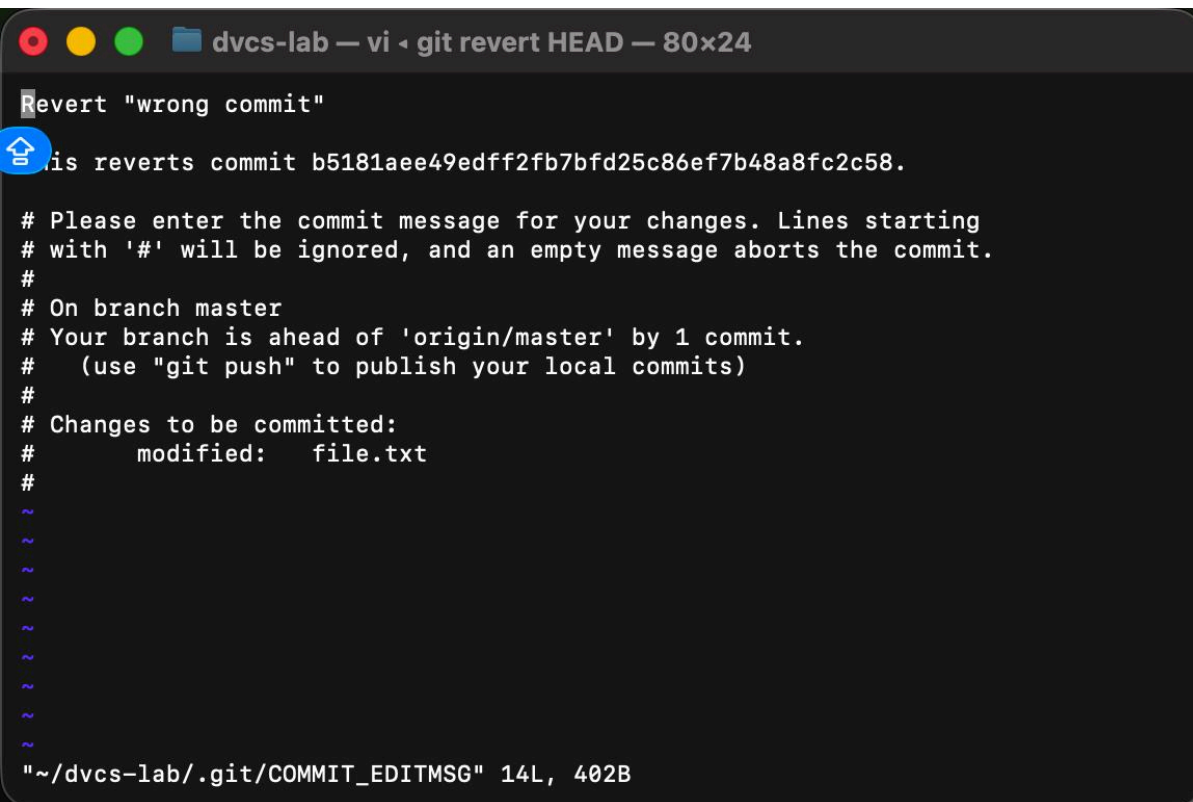
Task 6: Undo Changes using git revert

1. Introduce an incorrect change in file.txt and commit it.
2. Use git revert to undo the change.
3. Verify the file content and commit history.


```

[sheoraninfosec@Jigeshs-MacBook-Air ~ % cd dvcs-lab
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % echo "Wrong change" >> file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git add file.txt
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git commit -m "wrong commit"
[master b5181ae] wrong commit
1 file changed, 1 insertion(+)
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log --oneline
b5181ae (HEAD -> master) wrong commit
d95bd79 (origin/master, feature-branch) Added feature
f4048e7 Initial commit
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git revert HEAD
[master d94e186] Revert "wrong commit"
1 file changed, 1 deletion(-)
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %

```



```

Revert "wrong commit"
This reverts commit b5181ae49edff2fb7bfd25c86ef7b48a8fc2c58.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# Your branch is ahead of 'origin/master' by 1 commit.
# (use "git push" to publish your local commits)
#
# Changes to be committed:
#   modified:   file.txt
#
~
~
~
~
~
~
~
~/dvcs-lab/.git/COMMIT_EDITMSG 14L, 402B

```

Explanation

- Creates a new commit that undoes previous changes
- Safe for collaboration (history is preserved)
- Use **revert** in shared/public repositories

Task 7: Undo Changes using git reset

Perform the following operations and observe their effects:

(a) Soft Reset

- Undo the last commit using --soft

```

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git reset --soft HEAD~1
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log
commit b5181aee49edff2fb7bfd25c86ef7b48a8fc2c58 (HEAD -> master)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Thu Mar 26 11:14:32 2026 +0530

    wrong commit

commit d95bd79a1cbbba9225d1946ef980e5cbd70da66b7 (origin/master, feature-branch)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:55:27 2026 +0530

    Added feature

commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:40:37 2026 +0530

    Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %

```

Observation

1. Commit removed
2. Changes still staged

(b) Mixed Reset

- Undo another commit using default reset

```

[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git reset HEAD~1
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log
commit d95bd79a1cbbba9225d1946ef980e5cbd70da66b7 (HEAD -> master, origin/master, feature-branch)
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:55:27 2026 +0530

    Added feature

commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78
Author: sheoraninfosec <jsheoran9755@gmail.com>
Date: Wed Mar 25 09:40:37 2026 +0530

    Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %

```

Observation

1. Commit removed
2. Changes unstaged but still in working directory.

(c) Hard Reset

- Undo a commit using --hard

```
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git reset --hard HEAD~1  
HEAD is now at f4048e7 Initial commit  
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log  
commit f4048e7e20cb431d4652a3796ebcf9a7fedc9c78 (HEAD -> master)  
Author: sheoraninfosec <jsheoran9755@gmail.com>  
Date: Wed Mar 25 09:40:37 2026 +0530  
  
Initial commit  
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % █
```

Observation

- 1. Commit removed
- 2. Changes completely deleted

Reset Type	Commit	Staging Area	Working Directory
Soft	removed	kept	kept
Mixed	removed	cleared	kept
Hard	removed	cleared	deleted

Task 8: Comparative Analysis

Git as DVCS

- Full repo copy on every system
- Offline commits possible
- Faster operations
- No central dependency

DVCS vs Centralized VCS

Feature	DVCS (Git)	Centralized VCS
Reliability	High (local copies)	Low (server failure = data loss)
Performance	Fast (local ops)	Slow (network dependent)
Collaboration	Flexible (branching, merging)	Limited

Revert vs Reset

Use Case	Command
Undo safely (shared repo)	git revert
Rewrite history (local only)	git reset

Task 9: Visualization

1. Display the commit graph using:

git log --graph --oneline --all

```
[sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab % git log --graph --oneline --all
* d95bd79 (origin/master, feature-branch) Added feature
* f4048e7 (HEAD -> master) Initial commit
sheoraninfosec@Jigeshs-MacBook-Air dvcs-lab %
```

2. Interpret the branching and merging structure.

Interpretation

1. Shows branch structure visually
2. Helps understand:
 - a. Merge points
 - b. Parallel development

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